

Moving around

Hold spacebar then click and drag to move around.

Find it



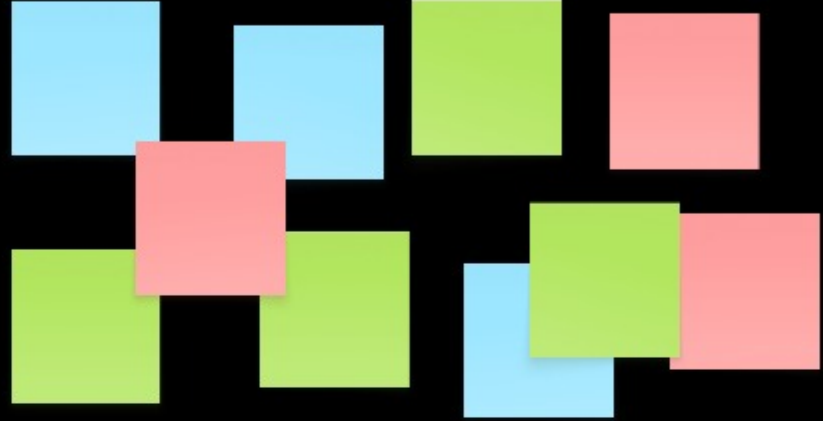
Hold **spacebar** then click and drag to move around

See it



Try it

- ① Hold **spacebar** until you see a hand
- ② Place your **cursor** over the stickies, then **click and drag**



 Edit the board when your cursor looks like a **pointer** (v). Move around with the **hand** (h).

Zooming in and out

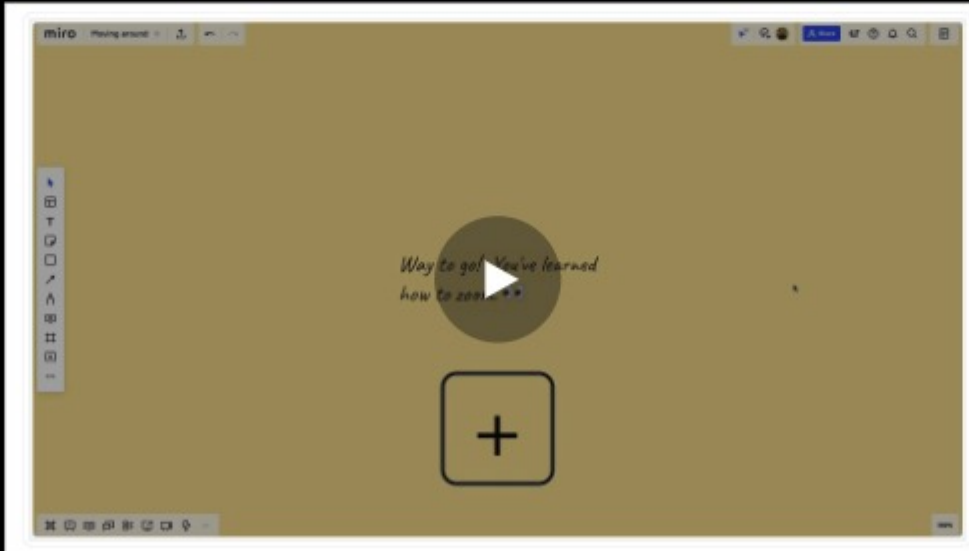
Use the plus and minus keys on your keyboard to zoom in and out.

Find it



Use the **plus** and **minus** keys on your keyboard

See it



Try it

- ① Center your screen on the small text below
- ② Use the **plus** and **minus** keys to zoom in and read the message

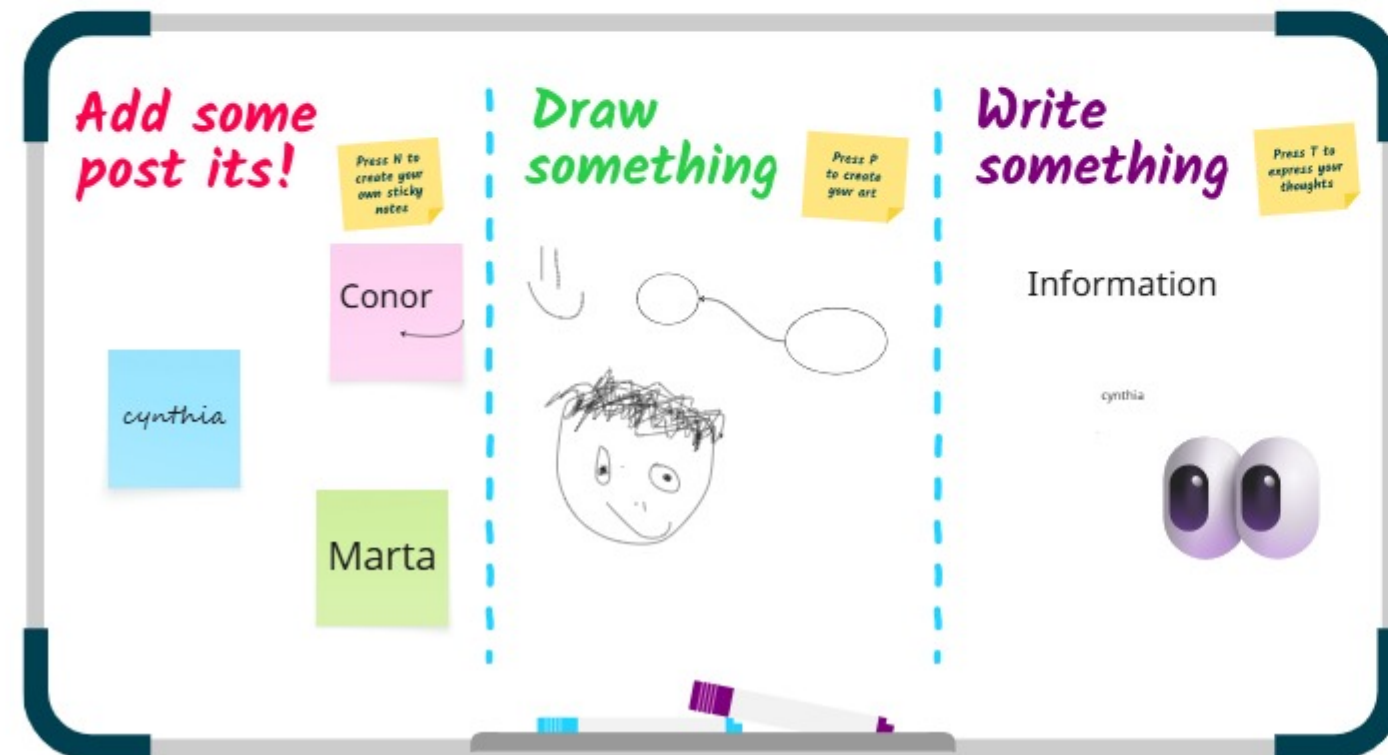


 Alternatively, use the **scroll wheel** on a mouse or **pinch** on a trackpad.

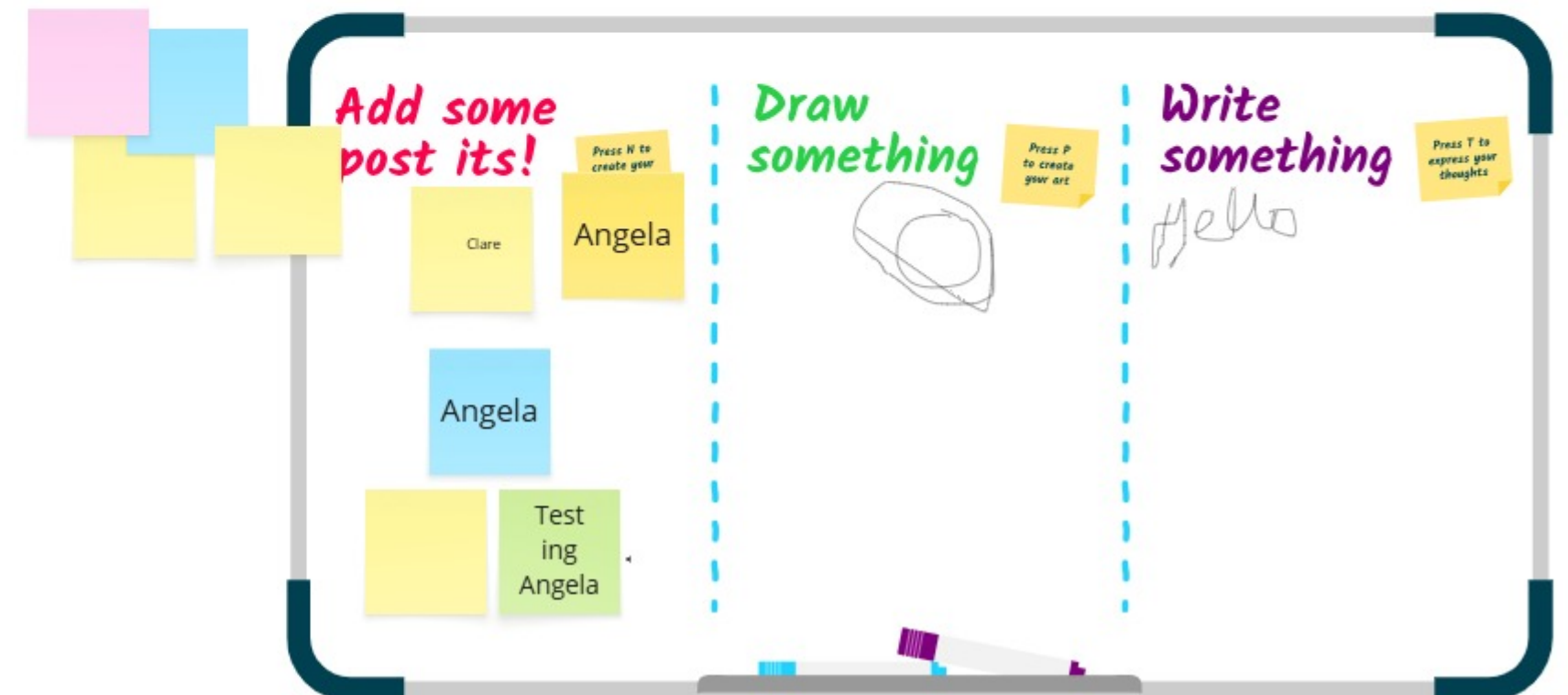
Have a play

Use the tools to get familiar with adding content.

Group 1



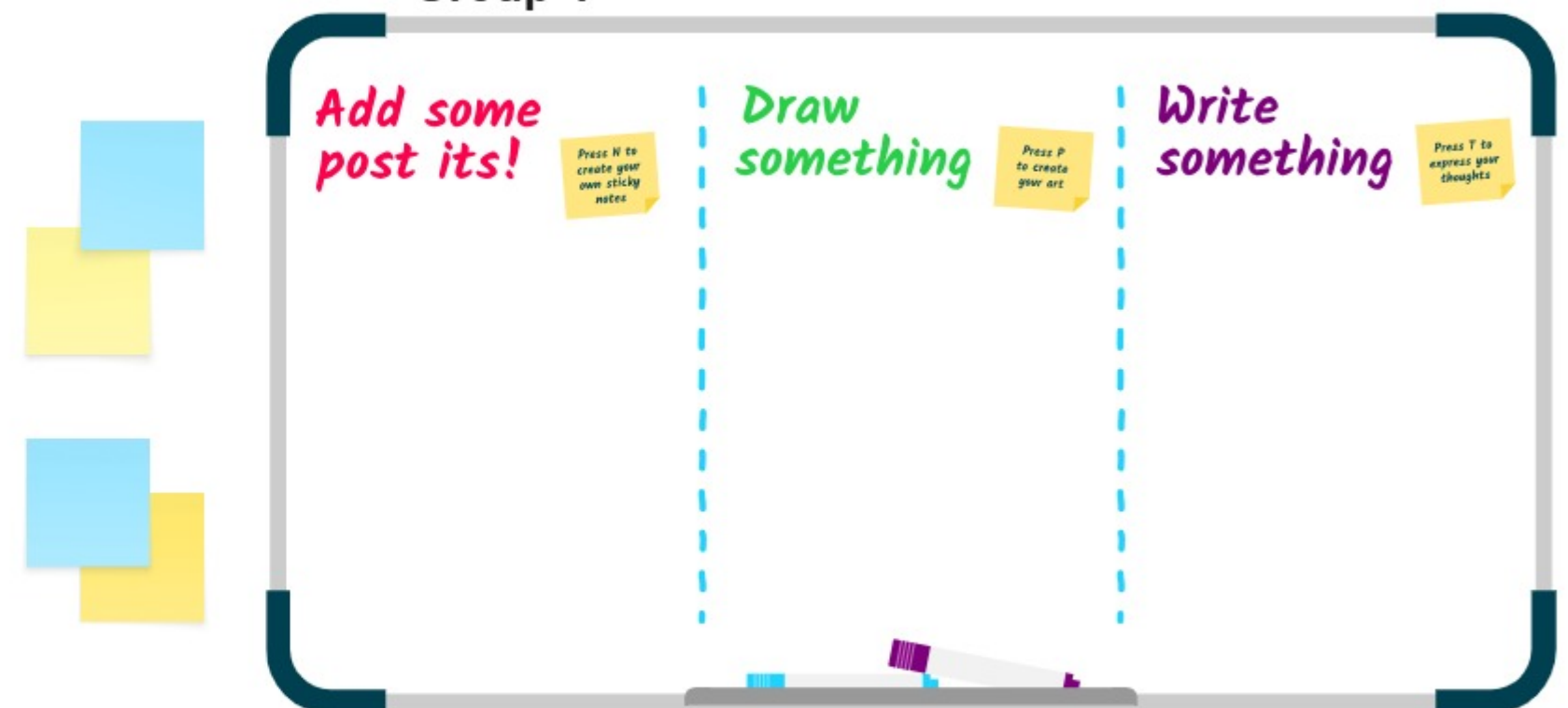
Group 2



Group 3



Group 4



Housekeeping

Rules regarding webcams

Keep your Webcam on throughout the delivery

Rules regarding mute

Keep your Microphone off throughout for privacy unless you are asked or need to respond.

Frequency of breaks

Short Breaks are scheduled every hour to support you with your learning.

Rules regarding group participation

Everyone should equally participate in Group discussion and take lead in contributing your view.

Rules regarding questions

The Golden Rule:

'There is no such thing as a stupid question'

'Even when asked by an instructor'

Please have a go at answering questions

Also:

'A question never resides in a single mind'
By asking questions, you are helping everyone out!

Anything else



Exercise

Entity 2

Task 1

Identify entities and provide entities with the information supplied.

There are at least 3 entities, needed and 2 of them are bridging tables.

1. Dealer
2. Motor
3. Motor/Dealer - Bridge
4. Motor
5. Motor/Dealer - Bridge

Task 2

Identify entities and provide entities with the information supplied.

There are at least 3 entities, needed and 2 of them are bridging tables.

1. Dealer
2. Motor
3. Motor/Dealer - Bridge
4. Motor
5. Motor/Dealer - Bridge

Task 3

Identify entities and provide entities with the information supplied.

There are at least 3 entities, needed and 2 of them are bridging tables.

1. Dealer
2. Motor
3. Motor/Dealer - Bridge
4. Motor
5. Motor/Dealer - Bridge

Task 4

Identify entities and provide entities with the information supplied.

There are at least 3 entities, needed and 2 of them are bridging tables.

1. Dealer
2. Motor
3. Motor/Dealer - Bridge
4. Motor
5. Motor/Dealer - Bridge

Task 5

Identify entities and provide entities with the information supplied.

There are at least 3 entities, needed and 2 of them are bridging tables.

1. Dealer
2. Motor
3. Motor/Dealer - Bridge
4. Motor
5. Motor/Dealer - Bridge

Task 6

Identify entities and provide entities with the information supplied.

There are at least 3 entities, needed and 2 of them are bridging tables.

1. Dealer
2. Motor
3. Motor/Dealer - Bridge
4. Motor
5. Motor/Dealer - Bridge

Task 7

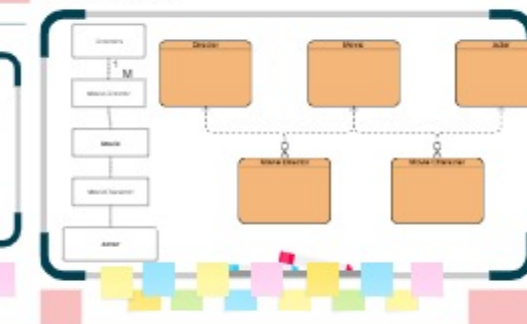
Identify entities and provide entities with the information supplied.

There are at least 3 entities, needed and 2 of them are bridging tables.

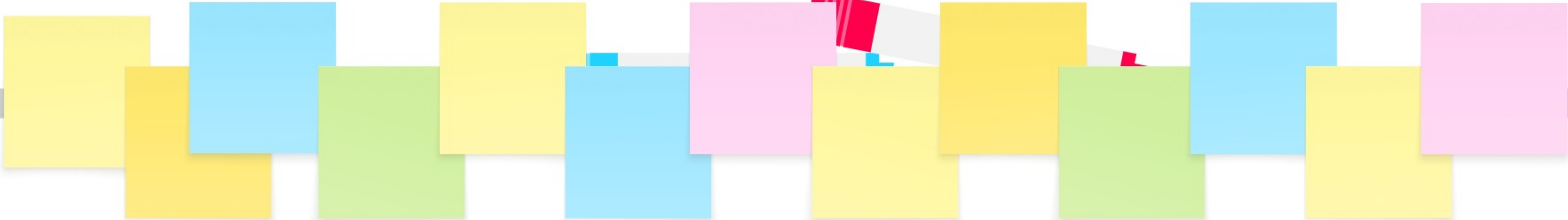
1. Dealer
2. Motor
3. Motor/Dealer - Bridge
4. Motor
5. Motor/Dealer - Bridge

Task 8

Identify entity relationships. Draw an **Entity Relationship Diagram** showing the relationships (cardinality) between the entities (tables) you identified in Task 1.



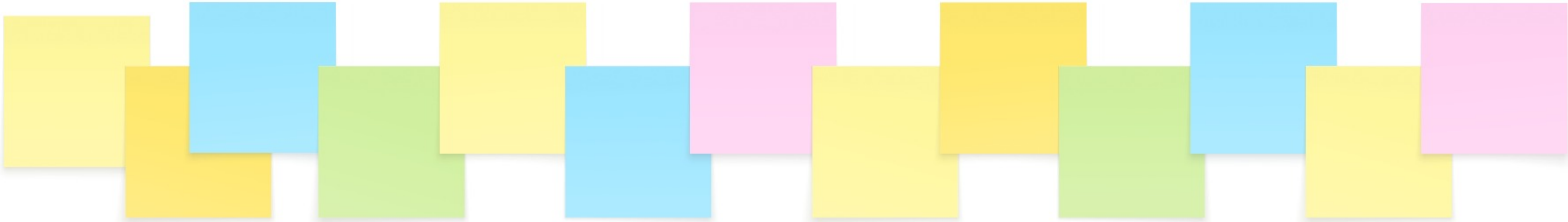
Entity description	Movies		
Database	Movie Database		
Entity name	Movies		
Key	Attribute	Data Type	Constraints/comments
PK	movie_id	INT	Not Null
	title	nvarchar(100)	Not Null
	budget	money	Null allowed
	overview	nvarchar(max)	Null allowed
	homepage	nvarchar(100)	Null allowed
	popularity	float	Null allowed
	release_date	date	Null allowed
	revenue	money	Null allowed
	runtime	INT	Null allowed
	movie_status	nvarchar(50)	Not Null
	tagline	nvarchar(max)	Null allowed
	vote_average	float	Null allowed
	vote_count	INT	Null allowed



Entity 3

Entity 1

Entity description	Table that acts as a bridging table between Movie and Director. Several directors can direct the same		
Database	Movie Database		
Entity name	DirectorMovie		
Key	Attribute	Data Type	Constraints/comments
FK	movie_id	int	Not Null
FK	director_id	int	Not Null



Teams

	Conor	Cynthia	Marta									
	Angela	Clare	Kris									
	Barry	Jim	Matthew									
												

Go to your team board

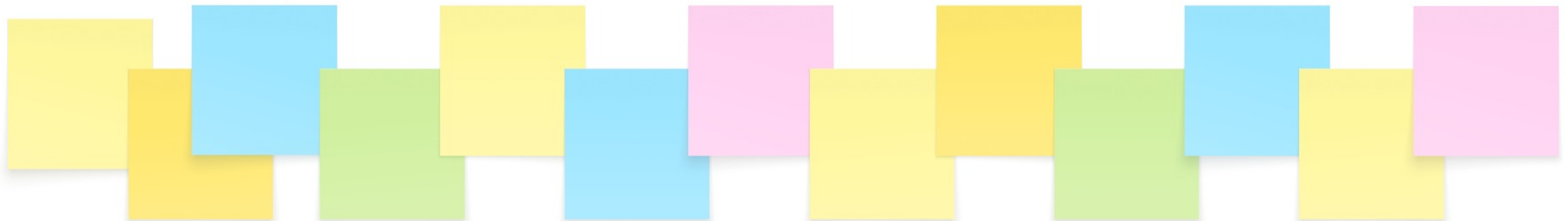
- 
Group 1
- 
Group 2
- 
Group 3
- 
Group 4

Task 1

Identify and name potential entities from the information supplied.

There are at least 5 entities needed and 2 of these are bridging tables.

1. Director
2. Movie
3. Movie Director - Bridge
4. Actor
5. MovieCharacter - Bridge

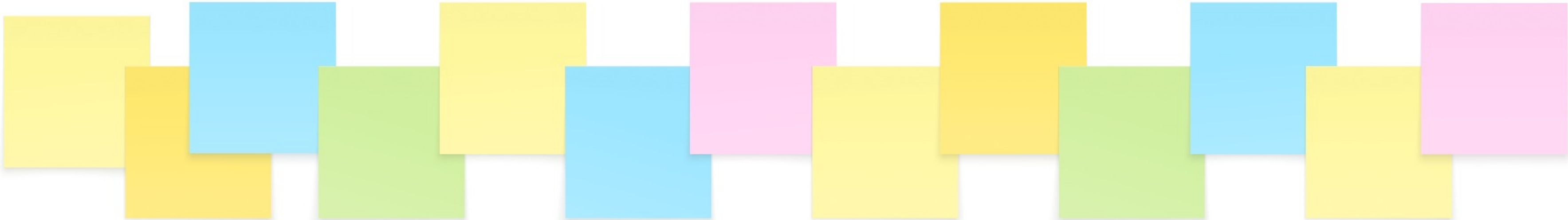


Movies and Actors Database

Director	Actor	gender	character_name	cast_order	movie_id	title	budget	homepage	overview	popularity	release_date	revenue	runtime	movie_status	tagline	vote_average	vote_count
Quentin Tarantino	Tim Roth	Male	Ted the Bellhop	43	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hand	8	65
Robert Rodriguez	Tim Roth	Male	Ted the Bellhop	44	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hand	8	65
Allison Anders	Tim Roth	Male	Ted the Bellhop	45	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hand	8	65
Alexandre Rockwell	Tim Roth	Male	Ted the Bellhop	46	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hand	8	65
Quentin Tarantino	Antonio Banderas	Male	Man	48	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hand	8	65
Robert Rodriguez	Antonio Banderas	Male	Man	49	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hand	8	65
Allison Anders	Antonio Banderas	Male	Man	50	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hand	8	65
Alexandre Rockwell	Antonio Banderas	Male	Man	51	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hand	8	65
Quentin Tarantino	Jennifer Beals	Female	Angela	52	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hand	8	65
Robert Rodriguez	Jennifer Beals	Female	Angela	53	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hand	8	65
Allison Anders	Jennifer Beals	Female	Angela	54	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hand	8	65

The attributes in the Movies and Actors table are:

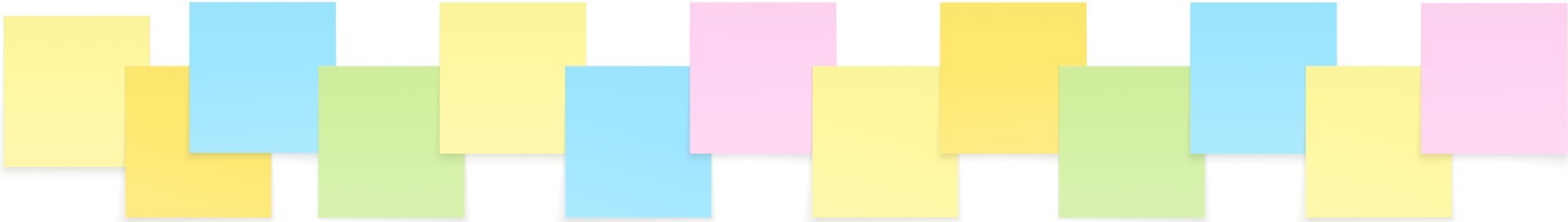
- Director
- Actor
- gender
- character_name
- cast_order
- movie_id
- title
- budget
- homepage
- overview
- popularity
- release_date
- revenue
- runtime
- movie_status
- tagline
- vote_average
- vote_count



Entity 5

Entity 1

Entity description	Table that acts as a bridging table between Movie and Character table.		
Database	Movie Database		
Entity name	MovieCharacter (bridging table)	Bridge	
Key	Attribute	Data Type	Constraints/comments
FK	movie_id	int	Not null
FK	actor_id	int	Not null
	character_id	int	Not null
	cast_order	int	Null allowed



Directors

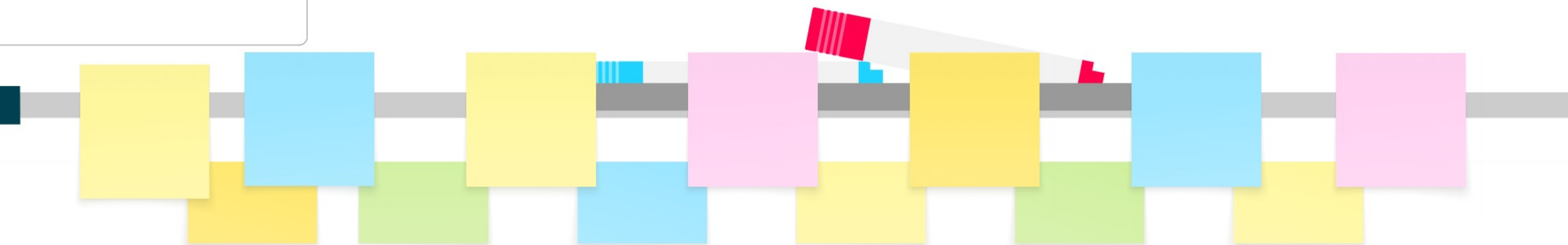
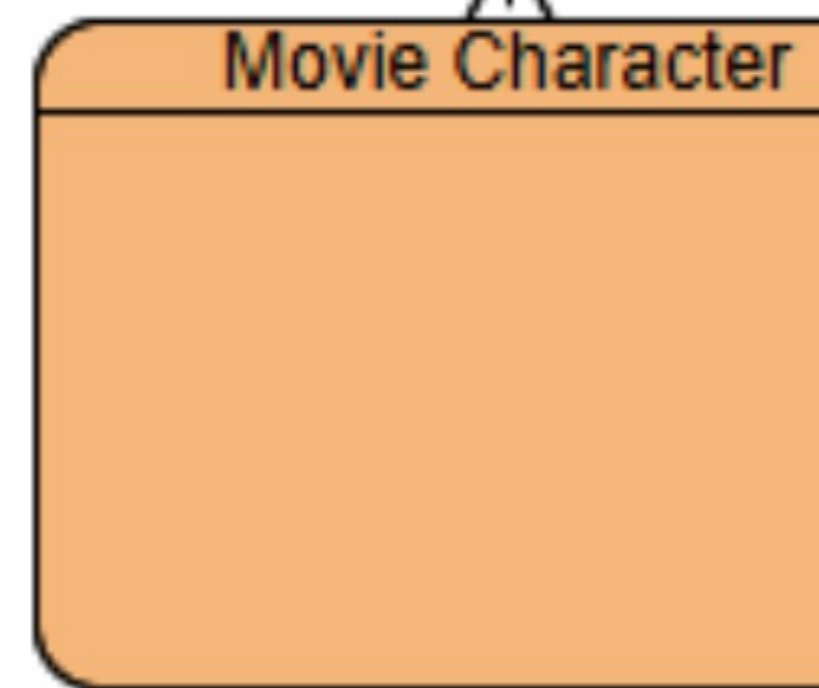
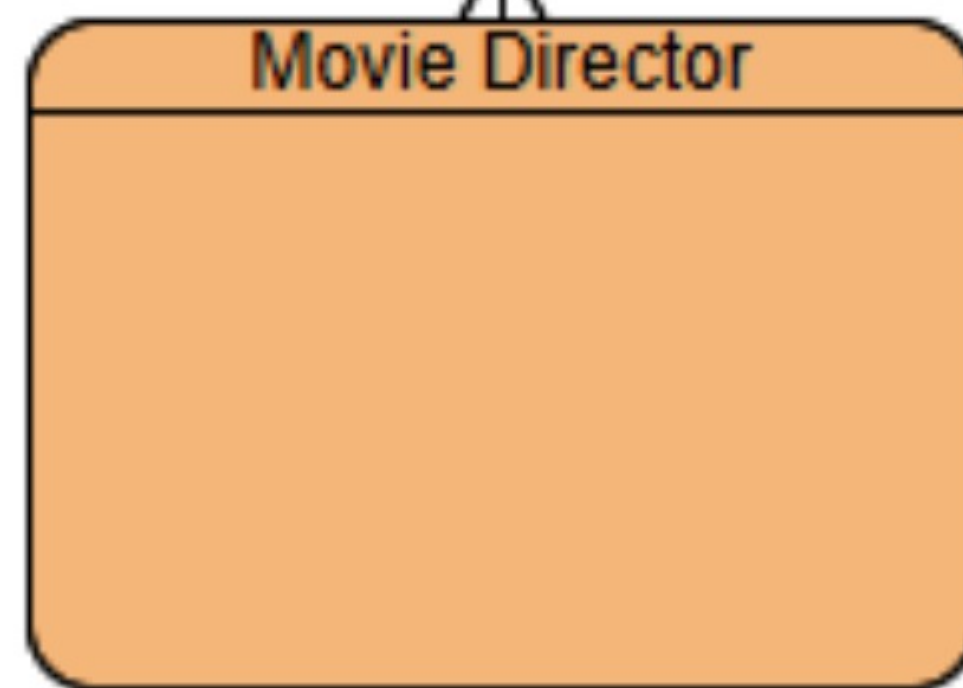
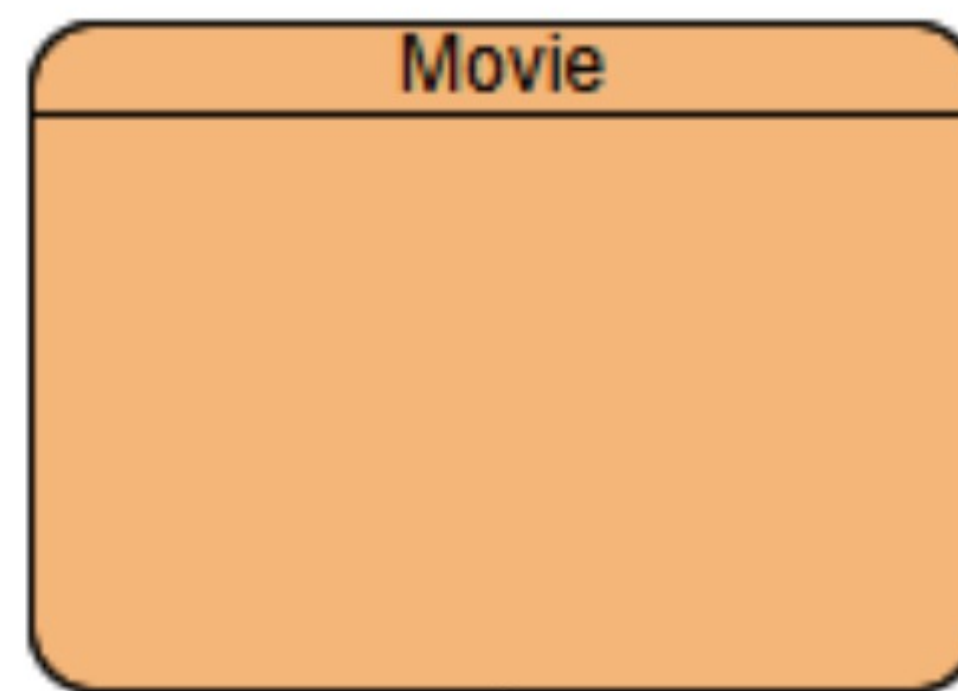
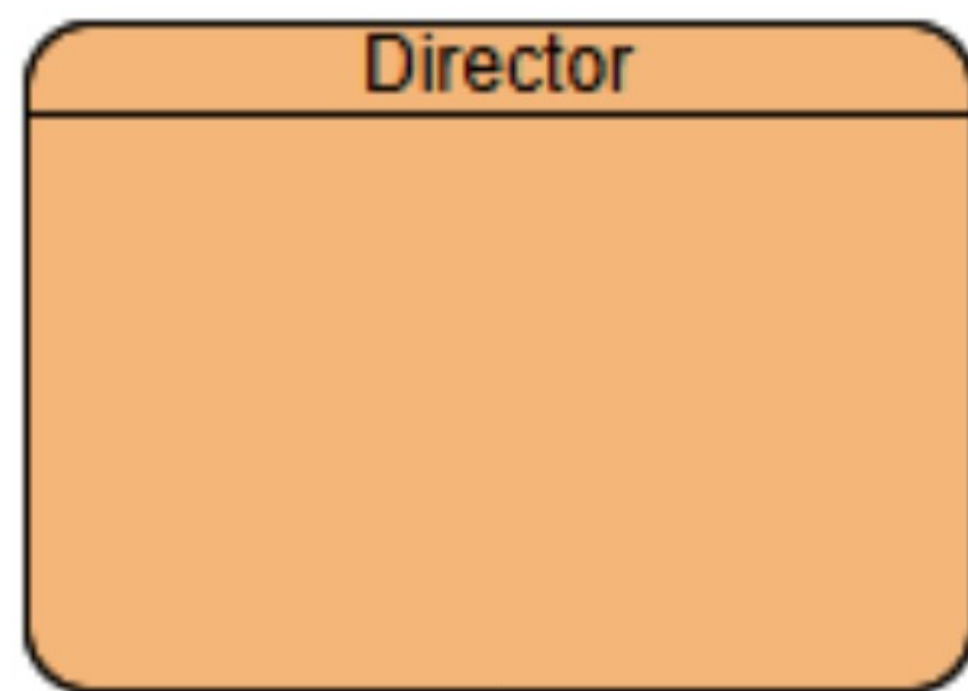
1
M

Movie.Director

Movie

MovieCharacter

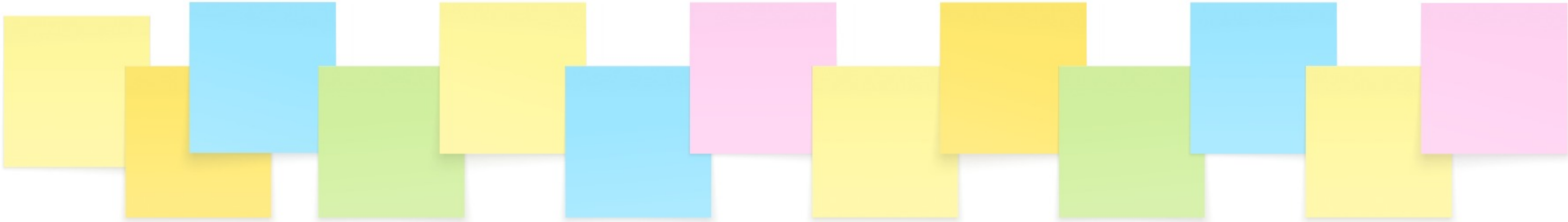
Actor



Entity 4

Entity 1

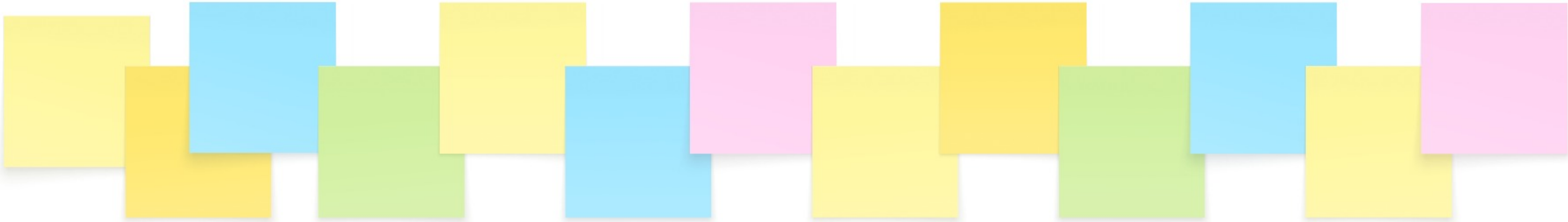
Entity description	Table containing details of actors recorded in the Movies database.		
Database	Movie Database		
Entity name	Actor		
Key	Attribute	Data Type	Constraints/comments
PK	actor_id	int	Not Null
	actor	nvarchar(100)	Null Allowed
	Gender	Boolean	Null



Task 2

Compile data dictionaries for your identified entities.
You may use the pro forma on the following page (or any other valid format).
You should include any Keys (Primary and Foreign), Data Types for each attribute and any suitable constraints I.e. Null, Not Null

Entity 1			
Entity description	Table that holds records of directors involved with the movies stored in the database.		
Database	Movie Database		
Entity name	Director		
Key	Attribute	Data Type	Constraints/comments
PK	director_id	INT	Not Nul
	director_name	nvarchar(100)	Not Null



The distribution of the features used in the model

Variable	N	p-Value
carat	1029	0.000000e+00
cut	1029	0.000000e+00
color	1029	0.000000e+00
clarity	1029	0.000000e+00
depth	1029	0.000000e+00
table	1029	0.000000e+00
x	1029	0.000000e+00
y	1029	0.000000e+00
z	1029	0.000000e+00
length	1029	0.000000e+00
width	1029	0.000000e+00
height	1029	0.000000e+00
weight	1029	0.000000e+00
volume	1029	0.000000e+00
price	1029	0.000000e+00

Task 1

Identify and mark a correct sentence from the information suggested.

There are at least 5 million, needed and 2 of these are leading ladies.

1. Director
2. Movie
3. actor
4. character
5. Movie Director



Longer data structures for your clients' profiles
You may choose your items and the following page to add their information.
You can search any type of property and it might take time to find details and any suitable resources in that time.

Key	Value	Key Type	Value Type
id	123456789	id	id
name	John Doe	name	name

Entity 2

ID	Address	House Type	How many rooms it has
01	12345 St.	det	3 rooms
02	67890 St.	det	3 rooms

Below the table is a small 3D house model and a row of 10 colored squares.

Category	Item	Value	Unit
Category 1	Item 1	100	Unit 1
Category 1	Item 2	200	Unit 1
Category 1	Item 3	300	Unit 1
Category 1	Item 4	400	Unit 1
Category 1	Item 5	500	Unit 1
Category 1	Item 6	600	Unit 1
Category 1	Item 7	700	Unit 1
Category 1	Item 8	800	Unit 1
Category 1	Item 9	900	Unit 1
Category 1	Item 10	1000	Unit 1
Category 1	Item 11	1100	Unit 1
Category 1	Item 12	1200	Unit 1
Category 1	Item 13	1300	Unit 1
Category 1	Item 14	1400	Unit 1
Category 1	Item 15	1500	Unit 1
Category 1	Item 16	1600	Unit 1
Category 1	Item 17	1700	Unit 1
Category 1	Item 18	1800	Unit 1
Category 1	Item 19	1900	Unit 1
Category 1	Item 20	2000	Unit 1
Category 1	Item 21	2100	Unit 1
Category 1	Item 22	2200	Unit 1
Category 1	Item 23	2300	Unit 1
Category 1	Item 24	2400	Unit 1
Category 1	Item 25	2500	Unit 1
Category 1	Item 26	2600	Unit 1
Category 1	Item 27	2700	Unit 1
Category 1	Item 28	2800	Unit 1
Category 1	Item 29	2900	Unit 1
Category 1	Item 30	3000	Unit 1
Category 1	Item 31	3100	Unit 1
Category 1	Item 32	3200	Unit 1
Category 1	Item 33	3300	Unit 1
Category 1	Item 34	3400	Unit 1
Category 1	Item 35	3500	Unit 1
Category 1	Item 36	3600	Unit 1
Category 1	Item 37	3700	Unit 1
Category 1	Item 38	3800	Unit 1
Category 1	Item 39	3900	Unit 1
Category 1	Item 40	4000	Unit 1
Category 1	Item 41	4100	Unit 1
Category 1	Item 42	4200	Unit 1
Category 1	Item 43	4300	Unit 1
Category 1	Item 44	4400	Unit 1
Category 1	Item 45	4500	Unit 1
Category 1	Item 46	4600	Unit 1
Category 1	Item 47	4700	Unit 1
Category 1	Item 48	4800	Unit 1
Category 1	Item 49	4900	Unit 1
Category 1	Item 50	5000	Unit 1
Category 1	Item 51	5100	Unit 1
Category 1	Item 52	5200	Unit 1
Category 1	Item 53	5300	Unit 1
Category 1	Item 54	5400	Unit 1
Category 1	Item 55	5500	Unit 1
Category 1	Item 56	5600	Unit 1
Category 1	Item 57	5700	Unit 1
Category 1	Item 58	5800	Unit 1
Category 1	Item 59	5900	Unit 1
Category 1	Item 60	6000	Unit 1
Category 1	Item 61	6100	Unit 1
Category 1	Item 62	6200	Unit 1
Category 1	Item 63	6300	Unit 1
Category 1	Item 64	6400	Unit 1
Category 1	Item 65	6500	Unit 1
Category 1	Item 66	6600	Unit 1
Category 1	Item 67	6700	Unit 1
Category 1	Item 68	6800	Unit 1
Category 1	Item 69	6900	Unit 1
Category 1	Item 70	7000	Unit 1
Category 1	Item 71	7100	Unit 1
Category 1	Item 72	7200	Unit 1
Category 1	Item 73	7300	Unit 1
Category 1	Item 74	7400	Unit 1
Category 1	Item 75	7500	Unit 1
Category 1	Item 76	7600	Unit 1
Category 1	Item 77	7700	Unit 1
Category 1	Item 78	7800	Unit 1
Category 1	Item 79	7900	Unit 1
Category 1	Item 80	8000	Unit 1
Category 1	Item 81	8100	Unit 1
Category 1	Item 82	8200	Unit 1
Category 1	Item 83	8300	Unit 1
Category 1	Item 84	8400	Unit 1
Category 1	Item 85	8500	Unit 1
Category 1	Item 86	8600	Unit 1
Category 1	Item 87	8700	Unit 1
Category 1	Item 88	8800	Unit 1
Category 1	Item 89	8900	Unit 1
Category 1	Item 90	9000	Unit 1
Category 1			

rally 2

Key	Address	Role	Type	Environment/Accession
01	vetera_01	car		Best Buy
	v-001	vanderloo (198)		Best Buy
	garden	vanderloo (198)		Best Buy

Strategy 1

Key	Attribute	Value Type	Value
10	Attribute 1	Integer	10
10	Attribute 2	String	10
10	Attribute 3	String	10
10	Attribute 4	String	10

Diagram below the table showing a sequence of colored blocks (yellow, blue, green, pink) and a red block, with a red arrow pointing to the red block.

07/01/2020

Consider study relationships: Study as a **very unstructured category** involving the relationships (temporally structured for contribution) you establish in Task 1

Study

Exercise

Work

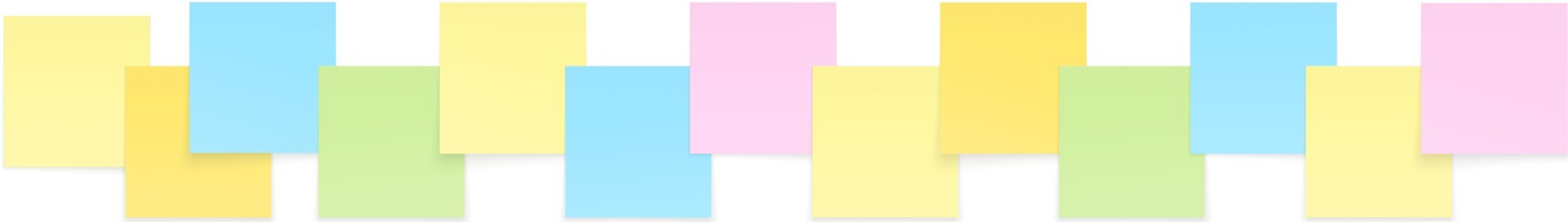
Exercise break

Commute

Entity 2

Entity 1

Entity description	Bridging table between Movie and Director		
Database	Movie		
Entity name	Direcor-Movie		
Key	Attribute	Data Type	Constraints/comments
FK	movie_id	int	Not Null
FK	director_id	int	Not Nuill



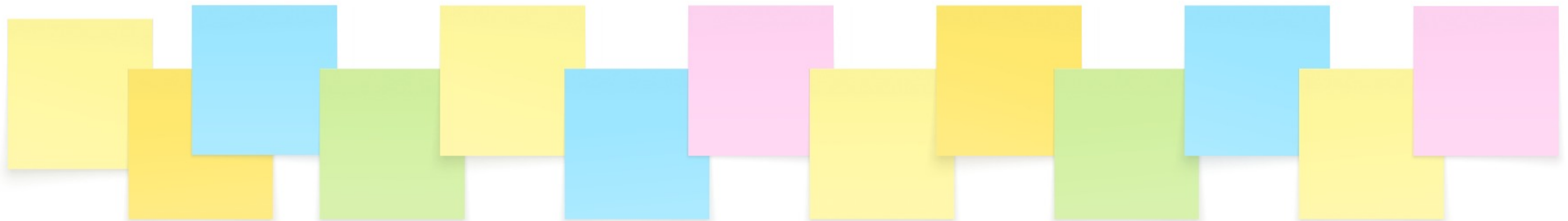
Entity description	Table with movie details		
Database	Movie		
Entity name	Movie		
Key	Attribute	Data Type	Constraints/comments
PK	MoveID	Int	Not null
FK	DirectorID	Int	Not null
	Title	nvarchar(100)	Not null
	budget	money	null
	homepage	nvarchar(150)	null
	overview	nvarchar(max)	null
	popularity	float	null
	release_date	date	null
	revenue	money	null
	runtime	int	null
	movie_status	boolean	null
	tagline	nvarchar(100)	null
	vote_average	float	null
	vote_count	int	null

Task 1

Identify and name potential entities from the information supplied.

There are at least 5 entities needed and 2 of these are bridging tables.

1. **Director**
2. **Movie**
3. actor
4. character
5. Movie-Director

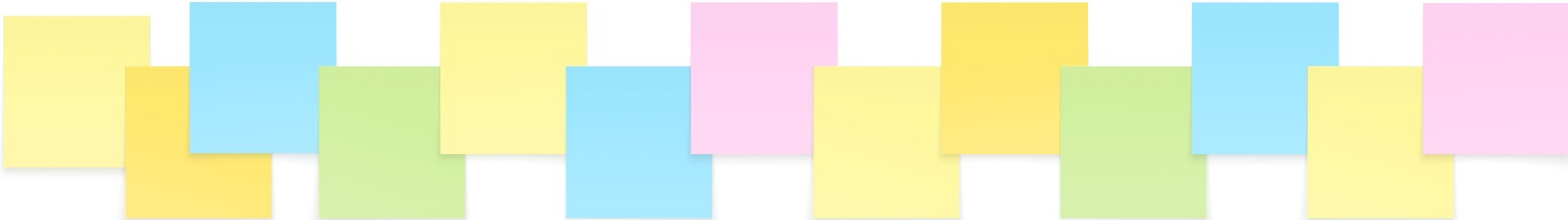


Movies and Actors Database

Director	Actor	gender	character_name	cast_order	movie_id	title	budget	homepage	overview	popularity	release_date	revenue	runtime	movie_status	tagline	vote_average	vote_count
Quentin Tarantino	Tim Roth	Male	Ted the Bellhop	43	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hand	8	65
Robert Rodriguez	Tim Roth	Male	Ted the Bellhop	44	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hand	8	65
Allison Anders	Tim Roth	Male	Ted the Bellhop	45	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hand	8	65
Alexandre Rockwell	Tim Roth	Male	Ted the Bellhop	46	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hand	8	65
Quentin Tarantino	Antonio Banderas	Male	Man	48	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hand	8	65
Robert Rodriguez	Antonio Banderas	Male	Man	49	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hand	8	65
Allison Anders	Antonio Banderas	Male	Man	50	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hand	8	65
Alexandre Rockwell	Antonio Banderas	Male	Man	51	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hand	8	65
Quentin Tarantino	Jennifer Beals	Female	Angela	52	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hand	8	65
Robert Rodriguez	Jennifer Beals	Female	Angela	53	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hand	8	65
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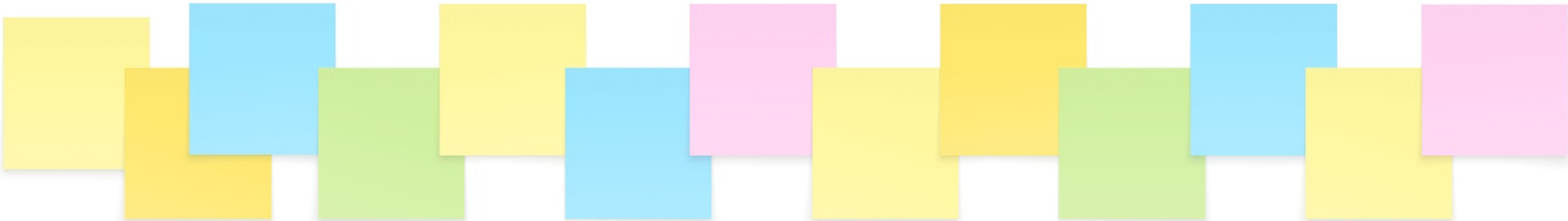
The attributes in the Movies and Actors table are:

- Director
- Actor
- gender
- character_name
- cast_order
- movie_id
- title
- budget
- homepage
- overview
- popularity
- release_date
- revenue
- runtime
- movie_status
- tagline
- vote_average
- vote_count



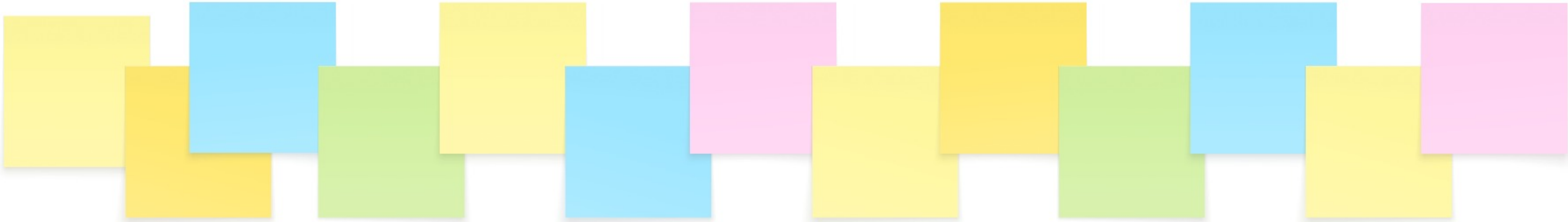
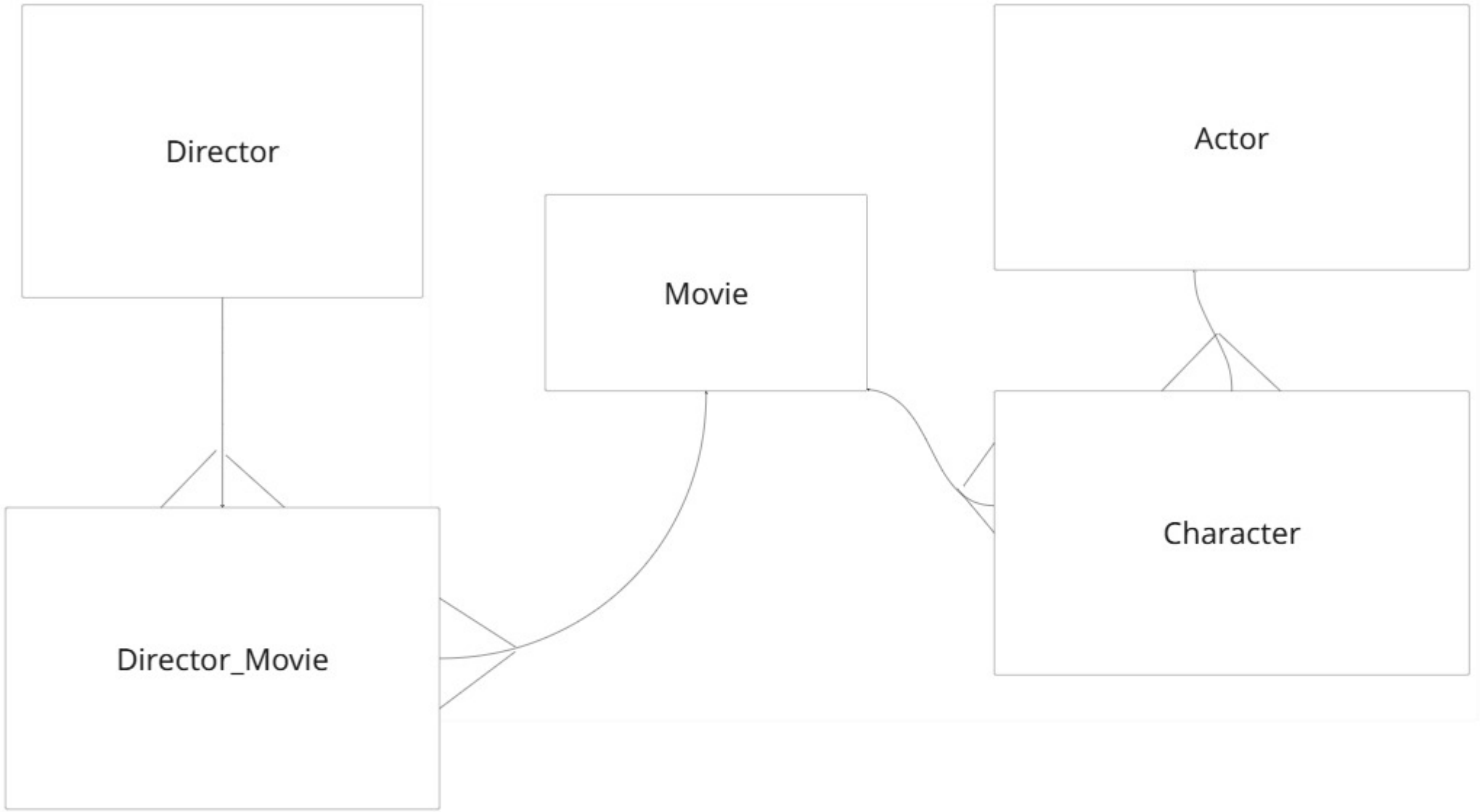
Entity 5

Entity description	Characters in movie		
Database	Movie		
Entity name	Character		
Key	Attribute	Data Type	Constraints/comments
	charactor name	nvarchar(150)	null
FK	movieID	int	not null
FK	actorID	int	not null
	cast_order	int	null



Task 3

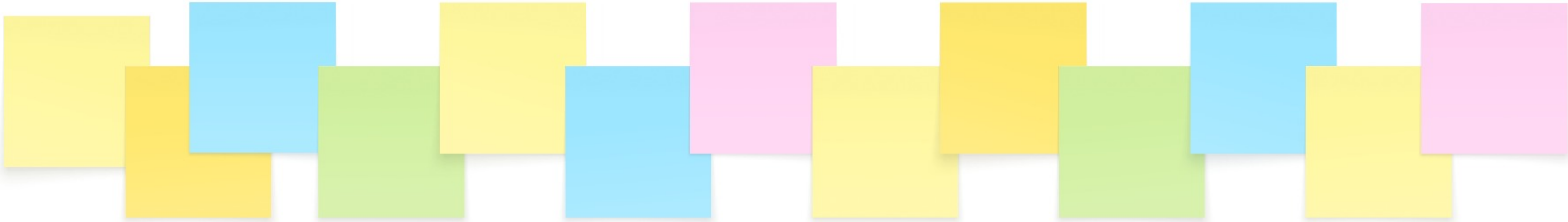
Identify entity relationships. Draw an **Entity Relationship Diagram** showing the relationships (cardinality) between the entities (tables) you identified in Task 1.



Entity 4

Entity 1

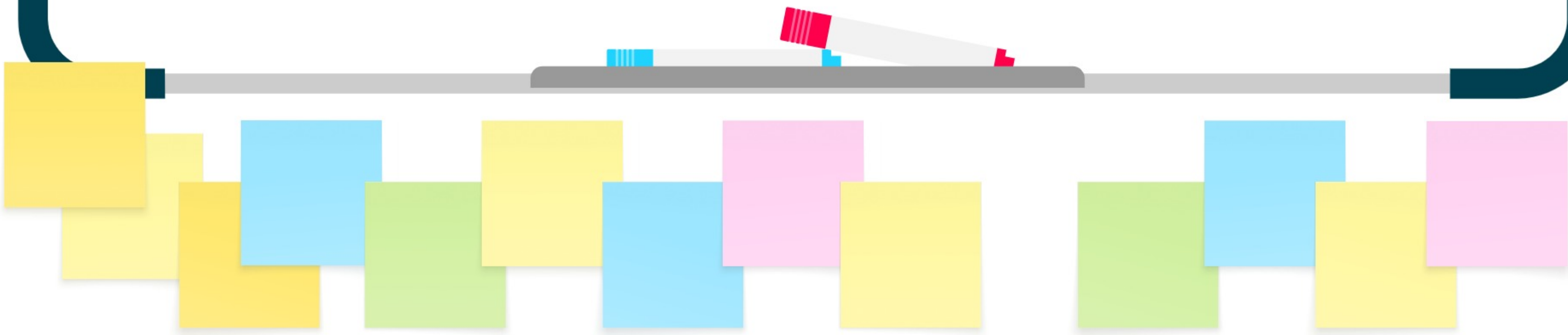
Entity description	Actor		
Database	Movies		
Entity name	Actor		
Key	Attribute	Data Type	Constraints/comments
PK	actor_id	int	Not Null
	actor	nvarchar (100)	Not Null
	gender	nvarchar (50)	Not Null



Task 2

Compile data dictionaries for your identified entities.
You may use the pro forma on the following page (or any other valid format).
You should include any Keys (Primary and Foreign), Data Types for each attribute and any suitable constraints I.e. Null, Not Null

Entity 1			
Entity description	Directors involved in the movie		
Database	Movies		
Entity name	Director		
Key	Attribute	Data Type	Constraints/comments
PK	DirectorID	Int	Not Null
	Director	vnarchar(50)	Not Null





Exercise

Step 1: Initial Situation

Scenario: A chess knight is on the board. The board is empty except for the knight. The knight is on the square d5.

Task: Identify the initial situation and the goal.

Initial Situation: A chess knight is on the square d5.

Goal: The knight is on the square d5.

Step 2: Problem Statement

Scenario: A chess knight is on the board. The board is empty except for the knight. The knight is on the square d5.

Task: Identify the problem statement and the goal.

Problem Statement: The knight is on the square d5.

Goal: The knight is on the square d5.

Step 3: Solution

Scenario: A chess knight is on the board. The board is empty except for the knight. The knight is on the square d5.

Task: Identify the solution and the goal.

Solution: The knight is on the square d5.

Goal: The knight is on the square d5.

Step 4: Evaluation

Scenario: A chess knight is on the board. The board is empty except for the knight. The knight is on the square d5.

Task: Identify the evaluation and the goal.

Evaluation: The knight is on the square d5.

Goal: The knight is on the square d5.

Step 5: Conclusion

Scenario: A chess knight is on the board. The board is empty except for the knight. The knight is on the square d5.

Task: Identify the conclusion and the goal.

Conclusion: The knight is on the square d5.

Goal: The knight is on the square d5.

Step 6: Summary

Scenario: A chess knight is on the board. The board is empty except for the knight. The knight is on the square d5.

Task: Identify the summary and the goal.

Summary: The knight is on the square d5.

Goal: The knight is on the square d5.

Step 7: Final Result

Scenario: A chess knight is on the board. The board is empty except for the knight. The knight is on the square d5.

Task: Identify the final result and the goal.

Final Result: The knight is on the square d5.

Goal: The knight is on the square d5.

Step 8: Final Evaluation

Scenario: A chess knight is on the board. The board is empty except for the knight. The knight is on the square d5.

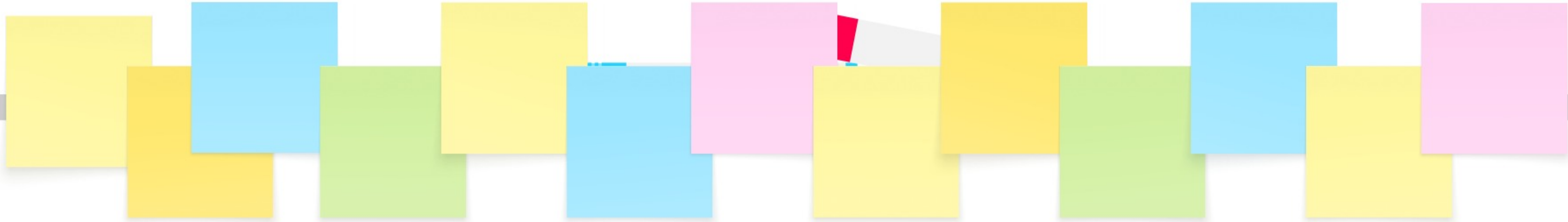
Task: Identify the final evaluation and the goal.

Final Evaluation: The knight is on the square d5.

Goal: The knight is on the square d5.

Entity 1

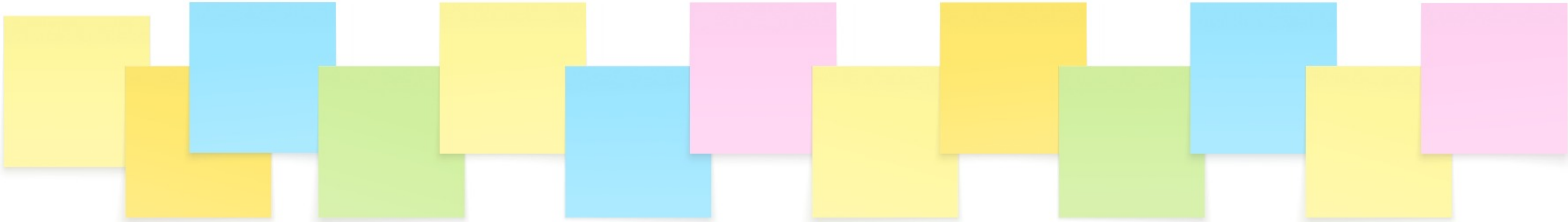
Entity description	Movies Table		
Database	Movies		
Entity name	Movies		
Key	Attribute	Data Type	Constraints/comments
PK	MovieID	Int	NOT NULL
	Title	nvarchar(100)	NOT NULL
	Popularity	float	NOT NULL
	tagline	nvarchar(max)	
	budget	money	NOT NULL
	vote_average	decimal(4,2)	
	release_date	Date	NOT NULL
	Runtime	Int	NOT NULL
	revenue	Money	NOT NULL
	homepage	nvarchar(150)	



Entity 3

Entity 1

Entity description	Actor Table		
Database	Movies		
Entity name	Actors		
Key	Attribute	Data Type	Constraints/comments
PK	ActorID	Int	NOT NULL
	Actor_name	nvarchar(100)	NOT NULL
	Gender	nvarchar(11)	

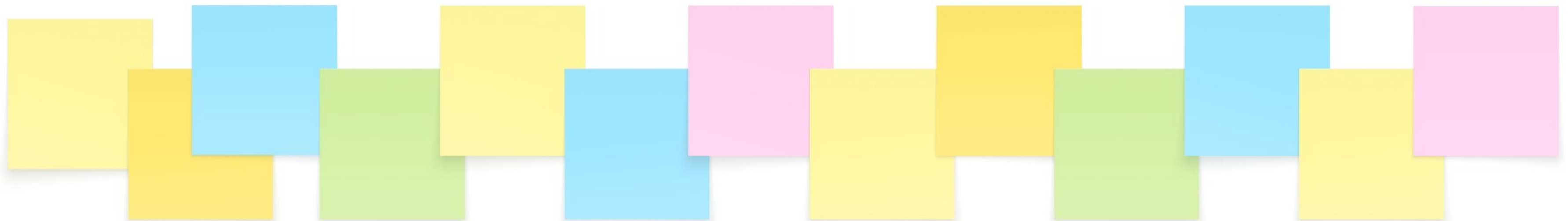


Task 1

Identify and name potential entities from the information supplied.

There are at least 5 entities needed and 2 of these are bridging tables.

1. **Movies**
2. **Directors**
3. **Actors**
4. **Characters**
5. **MovieDirector**

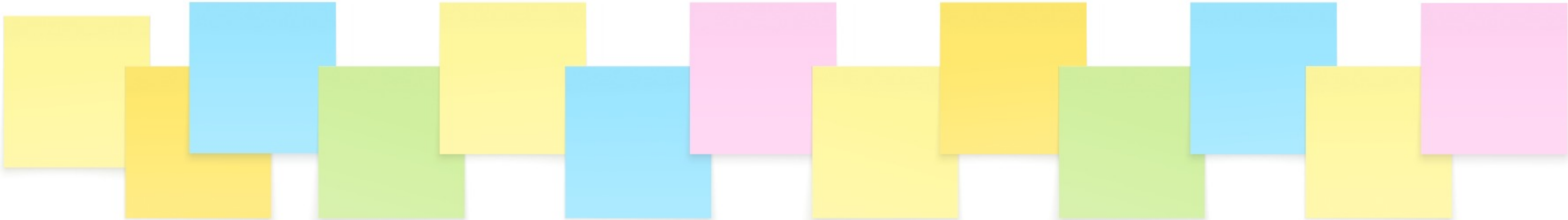


Movies and Actors Database

Director	Actor	gender	character_name	cast_order	movie_id	title	budget	homepage	overview	popularity	release_date	revenue	runtime	movie_status	tagline	vote_average	vote_count
Quentin Tarantino	Tim Roth	Male	Ted the Bellhop	43	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hand	8	65
Robert Rodriguez	Tim Roth	Male	Ted the Bellhop	44	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hand	8	65
Allison Anders	Tim Roth	Male	Ted the Bellhop	45	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hand	8	65
Alexandre Rockwell	Tim Roth	Male	Ted the Bellhop	46	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hand	8	65
Quentin Tarantino	Antonio Banderas	Male	Man	48	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hand	8	65
Robert Rodriguez	Antonio Banderas	Male	Man	49	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hand	8	65
Allison Anders	Antonio Banderas	Male	Man	50	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hand	8	65
Alexandre Rockwell	Antonio Banderas	Male	Man	51	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hand	8	65
Quentin Tarantino	Jennifer Beals	Female	Angela	52	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hand	8	65
Robert Rodriguez	Jennifer Beals	Female	Angela	53	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hand	8	65
Allison Anders	Jennifer Beals	Female	Angela	54	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hand	8	65

The attributes in the Movies and Actors table are:

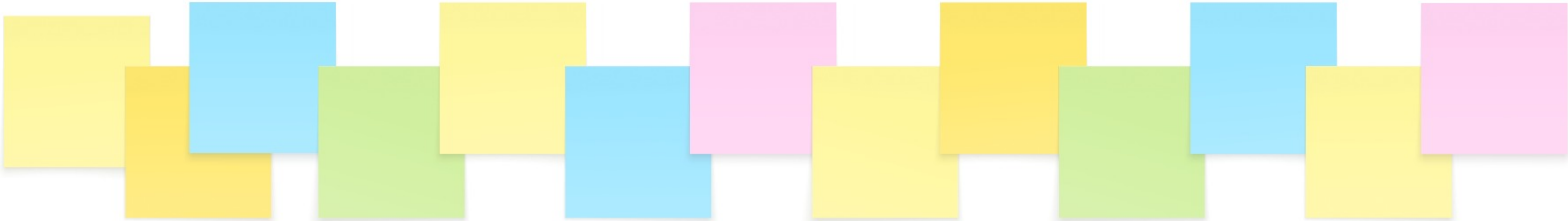
- Director
- Actor
- gender
- character_name
- cast_order
- movie_id
- title
- budget
- homepage
- overview
- popularity
- release_date
- revenue
- runtime
- movie_status
- tagline
- vote_average
- vote_count



Entity 5

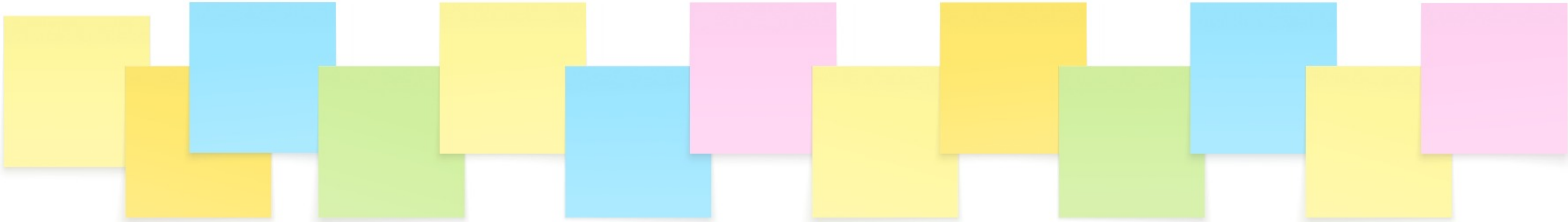
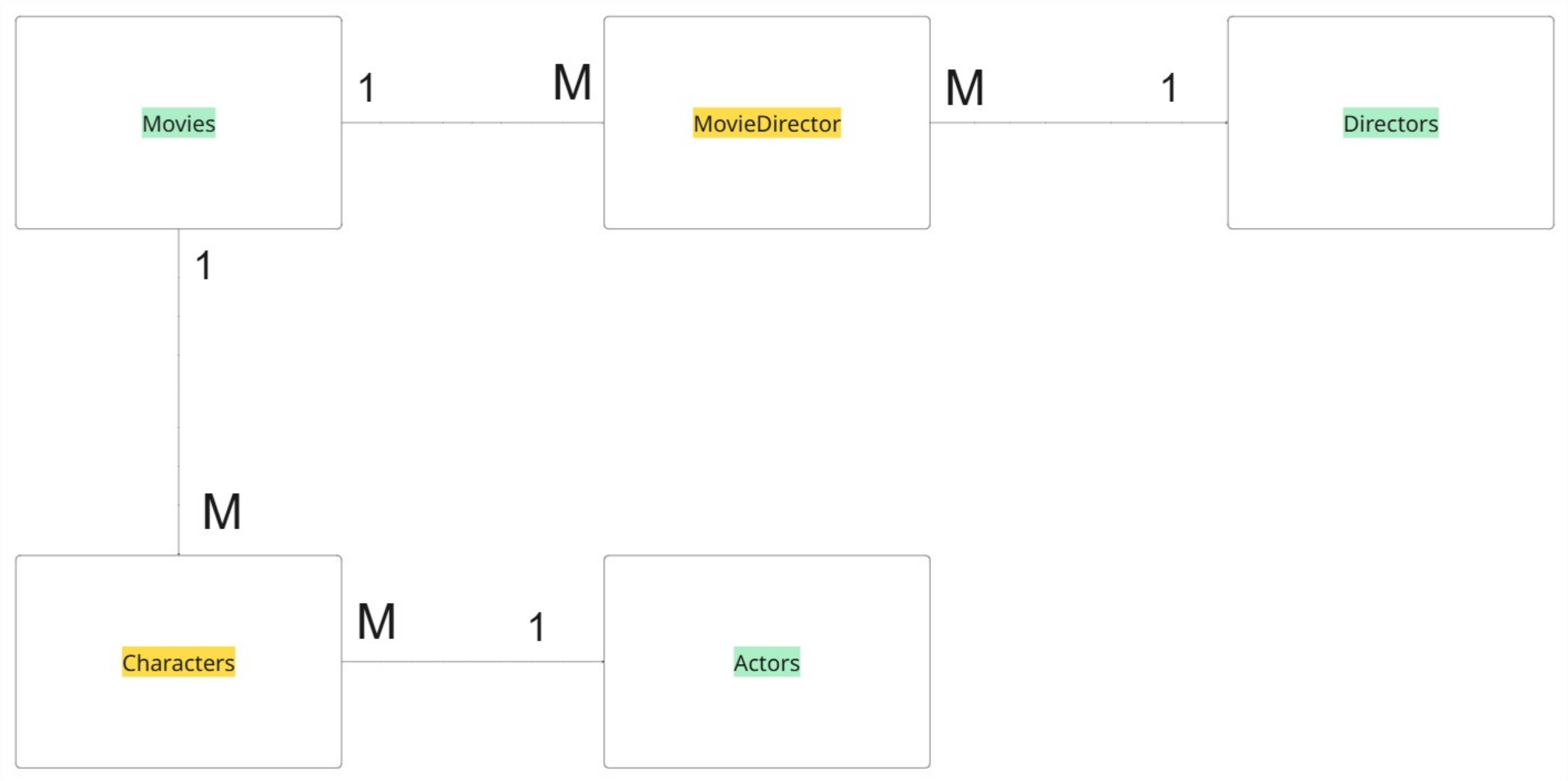
Entity 1

Entity description	MovieDirector Bridge		
Database	Movies		
Entity name	MovieDirector		
Key	Attribute	Data Type	Constraints/comments
FK	movie_id	Int	NOT NULL
FK	director_id	Int	NOT NULL



Task 3

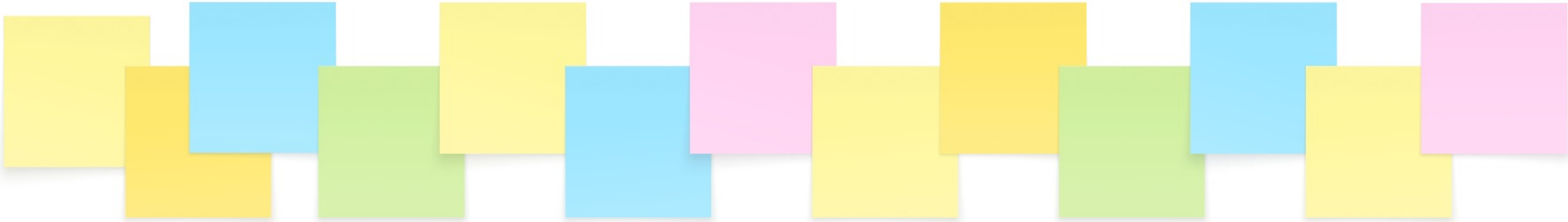
Identify entity relationships. Draw an **Entity Relationship Diagram** showing the relationships (cardinality) between the entities(tables) you identified in Task 1.



Entity 4

Entity 1

Entity description	Character Table		
Database	Movies		
Entity name	Charactera		
Key	Attribute	Data Type	Constraints/comments
FK	MovieID	Int	NOT NULL
FK	ActorsID	Int	NOT NULL
	cast_order	Int	NOT NULL
	Character_name	nvarchar(100)	NOT NULL

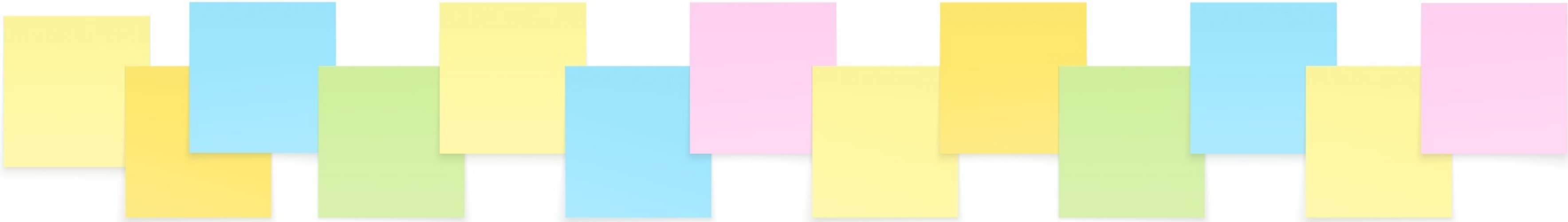


Task 2

Compile data dictionaries for your identified entities.
You may use the pro forma on the following page (or any other valid format).
You should include any Keys (Primary and Foreign), Data Types for each attribute and any suitable constraints I.e. Null, Not Null

Entity 1

Entity description	Directors Table		
Database	Movies		
Entity name	Directors		
Key	Attribute	Data Type	Constraints/comments
PK	DirectorsID	Int	NOT NULL
	Director	nvarchar(50)	NOT NULL





Exercise

[illegible]

Task 1

country, we were given no other further information supplied.

There are at least 8 entities, revealed and 2 of these are bridging entities.

1.
2.
3.
4.
5.



The screenshot shows a presentation slide titled "SPSS 1". The main content area displays a data management interface. On the left, there is a list of variables: "Country", "GDP", "Population", and "GDP per capita". The main area is a table with the following structure:

Country	GDP	Population	GDP per capita
USA	15000	300	50000
China	12000	1300	9000
India	6000	1100	5000
Germany	3000	80	35000
France	2000	60	30000
UK	2000	50	40000
Japan	4000	120	30000
Italy	2000	50	40000
Spain	1500	40	35000
Canada	2500	30	80000
Brazil	1000	180	5000
Argentina	500	30	15000
Mexico	1000	100	10000
South Africa	200	40	5000
Nigeria	100	150	500
Kenya	50	20	2000
India	50	10	500

Below the table, there is a bar chart showing the GDP per capita for each country. The x-axis represents the country, and the y-axis represents the GDP per capita. The bars are colored in a repeating pattern of yellow, blue, and green. A red arrow points to the bar for the USA, which is the highest.

The screenshot shows a digital workspace with a table and sticky notes. The table has the following structure:

Study description	Study type	Findings/Conclusions
Study name		
Key		

Below the table, there are several sticky notes in various colors (yellow, blue, green, pink) arranged in a row. Above the table, there are some small icons and a red arrow pointing to the table.

Strategy 1

Strategy	Address	Risk Type	Standard/Requirement

Timeline: [Bar with a red marker]

Sticky notes: [Yellow, Blue, Green, Yellow, Blue, Pink, Yellow, Green, Blue, Yellow, Pink]

2009-10

Entity	Attribute	Index Type	Index Space (bytes)
Entity 1	Attribute 1	Index Type 1	Index Space 1
Entity 1	Attribute 2	Index Type 2	Index Space 2
Entity 1	Attribute 3	Index Type 3	Index Space 3
Entity 2	Attribute 1	Index Type 1	Index Space 1
Entity 2	Attribute 2	Index Type 2	Index Space 2
Entity 2	Attribute 3	Index Type 3	Index Space 3
Entity 3	Attribute 1	Index Type 1	Index Space 1
Entity 3	Attribute 2	Index Type 2	Index Space 2
Entity 3	Attribute 3	Index Type 3	Index Space 3

Activity 2

Identify early relationships (links) and **key or steering projects** driving the relationships contributing to future the relationships you envision (see 1)

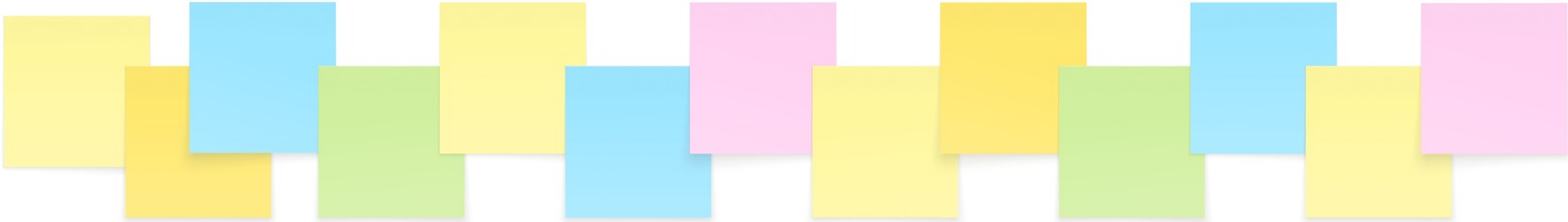


The diagram consists of a large, empty rectangular box with a thick blue border, representing a workspace for identifying relationships and projects. Below the box is a horizontal line with several small colored squares (yellow, green, blue, pink) and a red triangle, suggesting a timeline or sequence of events.

Entity 2

Entity 1

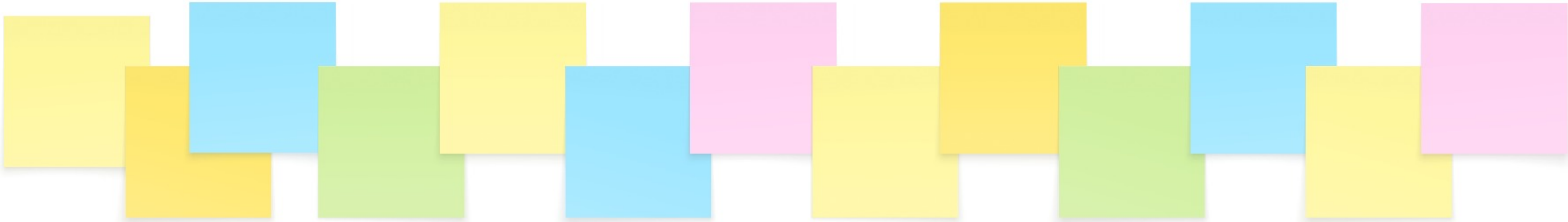
Entity description			
Database			
Entity name			
Key	Attribute	Data Type	Constraints/comments



Entity 3

Entity 1

Entity description			
Database			
Entity name			
Key	Attribute	Data Type	Constraints/comments

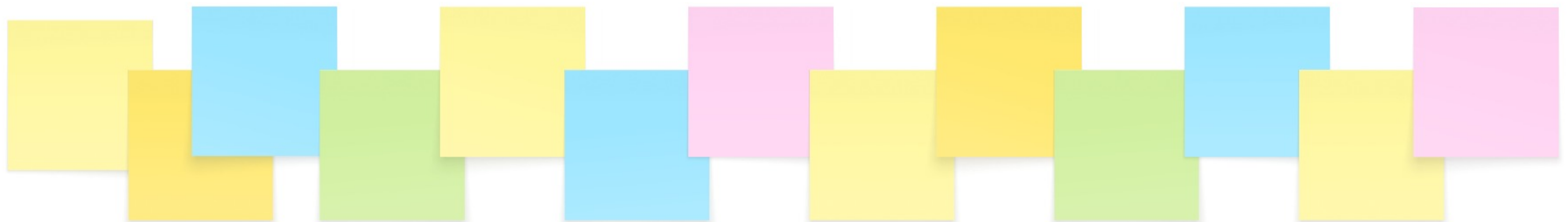


Task 1

Identify and name potential entities from the information supplied.

There are at least 5 entities needed and 2 of these are bridging tables.

1.
2.
3.
4.
5.

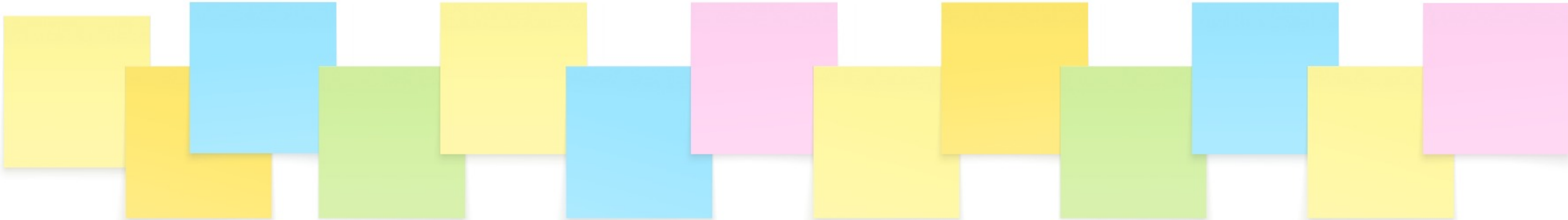


Movies and Actors Database

Director	Actor	gender	character_name	cast_order	movie_id	title	budget	homepage	overview	popularity	release_date	revenue	runtime	movie_status	tagline	vote_average	vote_count
Quentin Tarantino	Tim Roth	Male	Ted the Bellhop	43	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hand	8	65
Robert Rodriguez	Tim Roth	Male	Ted the Bellhop	44	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hand	8	65
Allison Anders	Tim Roth	Male	Ted the Bellhop	45	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hand	8	65
Alexandre Rockwell	Tim Roth	Male	Ted the Bellhop	46	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hand	8	65
Quentin Tarantino	Antonio Banderas	Male	Man	48	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hand	8	65
Robert Rodriguez	Antonio Banderas	Male	Man	49	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hand	8	65
Allison Anders	Antonio Banderas	Male	Man	50	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hand	8	65
Alexandre Rockwell	Antonio Banderas	Male	Man	51	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hand	8	65
Quentin Tarantino	Jennifer Beals	Female	Angela	52	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hand	8	65
Robert Rodriguez	Jennifer Beals	Female	Angela	53	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hand	8	65
Allison Anders	Jennifer Beals	Female	Angela	54	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hand	8	65

The attributes in the Movies and Actors table are:

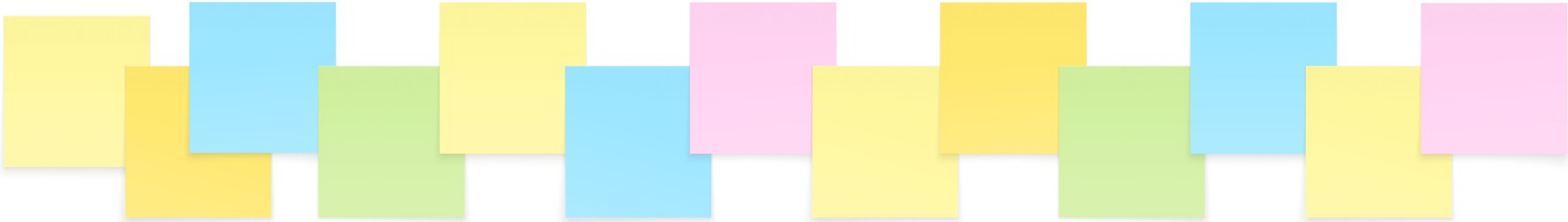
- Director
- Actor
- gender
- character_name
- cast_order
- movie_id
- title
- budget
- homepage
- overview
- popularity
- release_date
- revenue
- runtime
- movie_status
- tagline
- vote_average
- vote_count



Entity 5

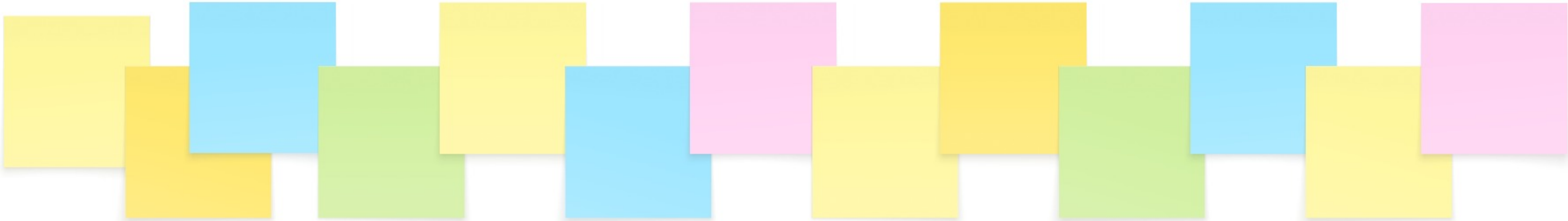
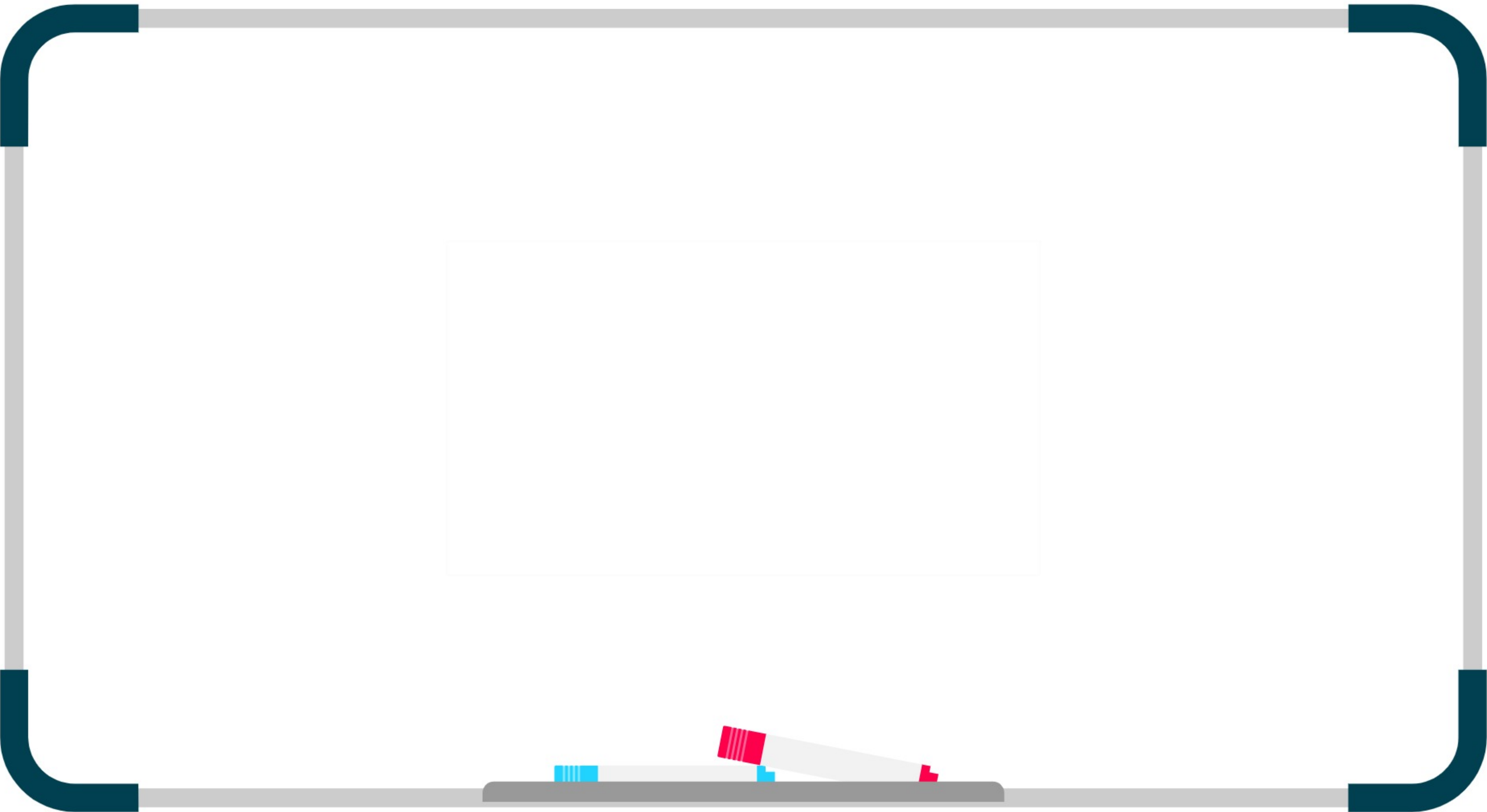
Entity 1

Entity description			
Database			
Entity name			
Key	Attribute	Data Type	Constraints/comments



Task 3

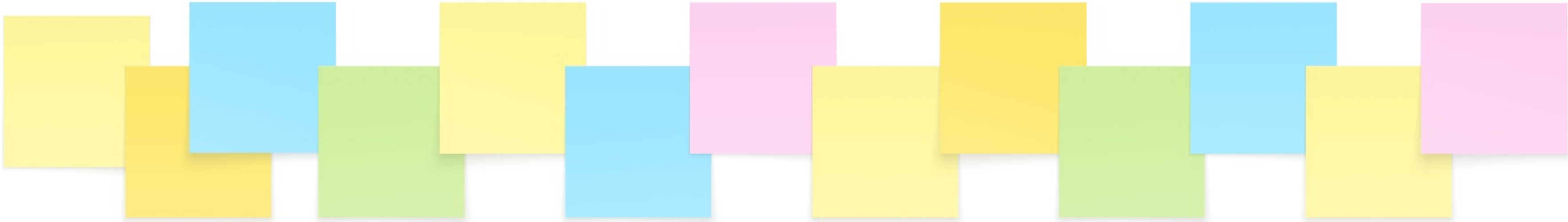
Identify entity relationships. Draw an **Entity Relationship Diagram** showing the relationships (cardinality) between the entities(tables) you identified in Task 1.



Entity 4

Entity 1

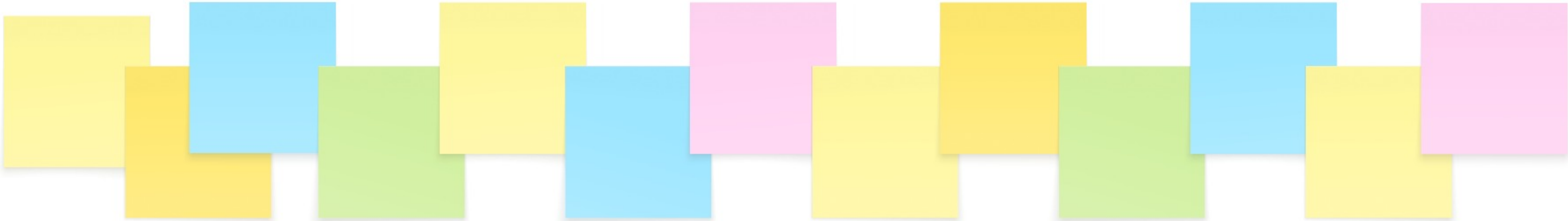
Entity description			
Database			
Entity name			
Key	Attribute	Data Type	Constraints/comments



Task 2

Compile data dictionaries for your identified entities.
You may use the pro forma on the following page (or any other valid format).
You should include any Keys (Primary and Foreign), Data Types for each attribute and any suitable constraints I.e. Null, Not Null

Entity 1			
Entity description			
Database			
Entity name			
Key	Attribute	Data Type	Constraints/comments



Activity 1: Movies Database Normalisation

Learner instructions

We have been given the task of creating an online resource based on movies produced in the last 50 years. One of the features of this resource will be a RDMS that will store details of the personnel and production information relating to a large number of films. We have an Excel file provided with data that has been partly normalised in that some of the entities and their attributes have already been identified. However one of the sheets on the Excel file labelled Movies and Actors still needs to be normalised before it is suitable for the RDMS. Screenshots can be seen below of this table and its attributes.

Task 1

The attributes in the Movies and Actors table are:

- Director
- Actor
- gender
- character_name
- cast_order
- movie_id
- title
- budget
- homepage
- overview
- popularity
- release_date
- revenue
- runtime
- movie_status
- tagline
- vote_average
- vote_count

Identify and name potential entities from the information supplied. Remember that there may be many-to-many relationships between the tables that would require a bridging table (a table containing the foreign keys of the tables to be joined) Think about the fact that and actor can play many characters but these characters can appear in a number of movies i.e. Luke Skywalker appears in Star Wars but also in The Empire Strikes Back! Also when you create any of the main tables from they will need a Primary Key attribute that may need to be added for example director_ID Identify and name potential entities from the information supplied. Hint: There are at least 5 entities needed and 2 of these are bridging tables.

1. Director
2.
3.
4.
5.

Task 2

Compile data dictionaries for your identified entities. You may use the pro forma on the following page (or any other valid format). You should include any Keys (Primary and Foreign), Data Types for each attribute and any suitable constraints I.e. Null, Not Null

Entity description			
Database			
Entity name			
Key	Attribute	Data Type	Constraints/comments

Task 3

Identify entity relationships. Draw an **Entity Relationship Diagram** showing the relationships (cardinality) between the entities(tables) you identified in Task 1. Show the Primary and Foreign keys to show how tables would relate to each other i.e. The Primary key of one table (Parent) would become the Foreign key in its related table (Child) Hint: Bridging tables that are used to avoid many-to-many relationships are basically formed of two one-to-many relationships. The diagram below shows the “many-to-many” relationship to avoid duplicate values or multiple values in the one column, between Orders and Products. OrderID and ProductID have become foreign keys in the OrderDetails table.

Replace the image above with your own Entity Relationship Diagram (ERD)
<https://online.visual-paradigm.com/>